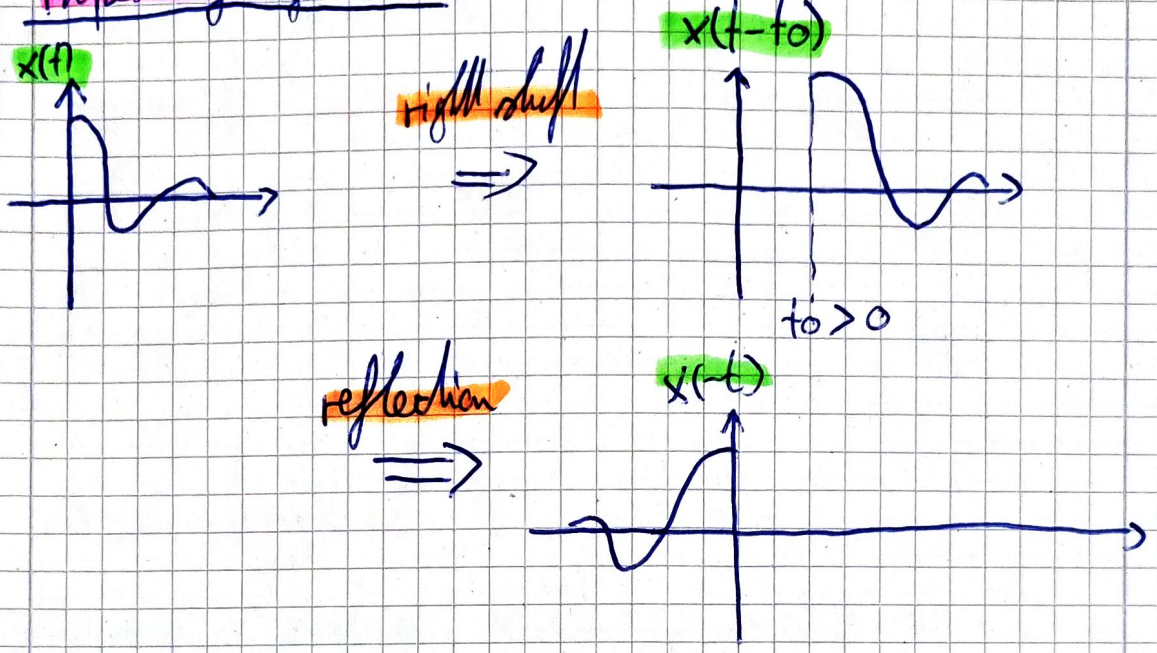
Signale & Systeme

Klassen von Signalen:

- analog: zeitkontinuierlich & Amplituden kontinuierlich
- discrete-time: zeit discrete & Amplitude kontinuierlich
- quantized: zeit kontinuierlich & Amplitude discrete
- digital: zeit discrete & Amplitude discrete

Properties of Signals:



Symetrie of Signals:

$$x(t) = x_e(t) + x_o(t)$$

even signal: $x_e(t) = x_e(-t)$; $x_e(t) = \frac{1}{2}[x(t) + x(-t)]$

odd signal: $x_o(t) = -x_o(-t)$; $x_o(t) = \frac{1}{2}[x(t) - x(-t)]$

$$x(t) = x_R(t) + j\,x_I(t)$$

$$x_R(t) = \frac{1}{2}[x(t) + x^*(t)]$$

$$x_I(t) = \frac{1}{2j}[x(t) - x^*(t)]$$

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Symmetrie of Complex Signals

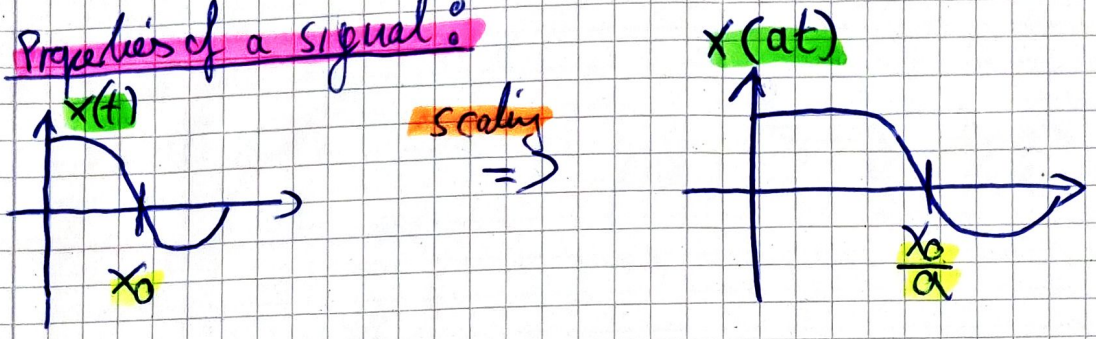
conjugate even signal:

$$x(t) = x^*(-t) \Rightarrow x_R(t) + j x_I(t) = x_R(-t) - j x_I(-t)$$

conjugate odd signal:

$$x(t) = -x^*(t) \Rightarrow x_R(t) + j x_I(t) = -x_R(-t) + j x_I(t)$$

Properties of a signal:



causal signals:

causal signal: $x(t) = 0$ for $t < 0$

anti-causal signal: $x(t) = 0$ for $t \geq 0$

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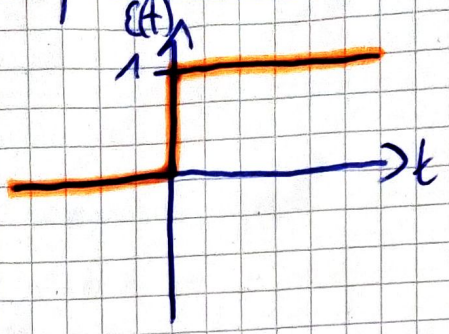
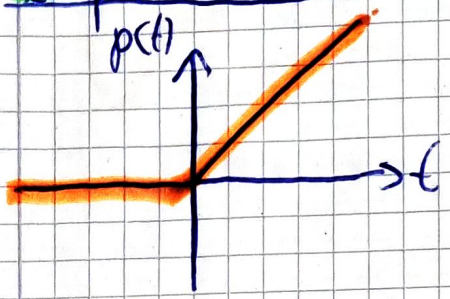
Elementary Signals

Step Function:



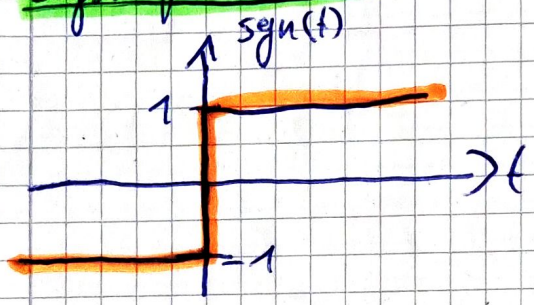
$$\varepsilon(t) = \begin{cases} 0, & t<0 \\ 1/2, & t=0 \\ 1, & t>1 \end{cases}$$

Ramp Function:



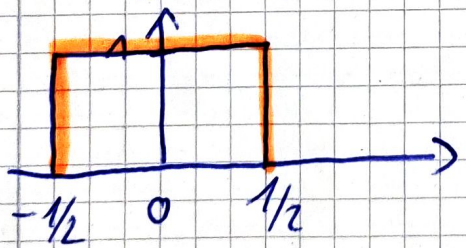
$$p(t) = x \cdot \varepsilon(t)$$

sign function:



$$sgn(t) = \begin{cases} -1, & t<0 \\ 0, & t=0 \\ 1, & t>0 \end{cases}$$

rectangular function:



$$rect(t) = \begin{cases} 0, & |t| > \frac{1}{2} \\ 1/2, & |t| = 1/2 \\ 1, & |t| < 1/2 \end{cases}$$

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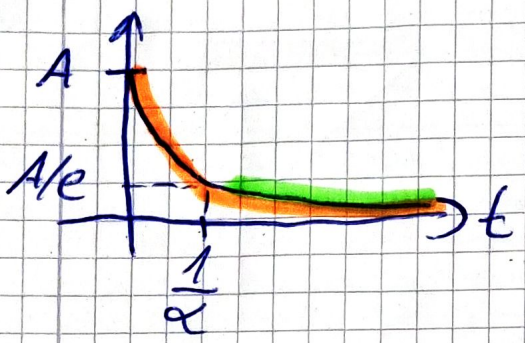
Elementary Signals

Triangular Function:

$$\Delta(t) = \begin{cases} 0, & |t| > 1 \\ 1-t, & 0 \le t \le 1 \\ 1+t, & -1 \le t \le 0 \end{cases}$$

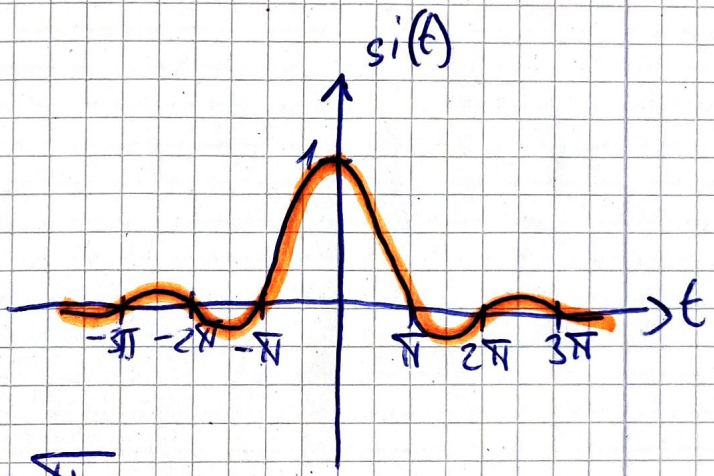
Real Exponential Function



$$x(t) = A e^{\alpha t}$$

Si-Function:

$$si(t) = \frac{\sin(t)}{t}$$



$$\int_{-\infty}^{\infty} si(t)\,dt = \int_{-\infty}^{\infty} si^2(t)\,dt = \pi$$

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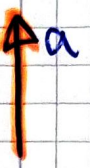
Dirac - Impulse

$$\delta(t) = \begin{cases} \neq 0 & ; t = 0 \\ = 0 & ; \text{sonst} \end{cases}$$

$$\int_{-\infty}^{\infty} \delta(t)\, dt = 1$$

$$\int_{-\infty}^{\infty} a\, \delta(t)\, dt = a$$

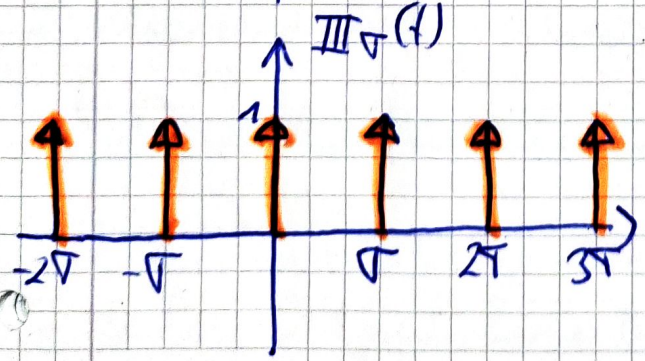


$$x(t)\,\delta(t-t_0) = x(t_0)\,\delta(t-t_0)$$

Other Scaling of the dirac:

$$\delta(at) = \frac{1}{|a|}\,\delta(t)$$

Dirac Impulse Comb



$$\text{III}_T(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT)$$

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Signale & Systeme

Power & Energie of a signal

continuous	discrete
Dimension of a signal	
$\dim\{x(t)\} = \sqrt{\text{power}}$	$\dim\{x[k]\} = \sqrt{\text{power}}$
Instantaneous power of a signal	
$P_{xi} = \lvert x(t)\rvert^2$	$P_{xi} = \lvert x[k]\rvert^2$
Power of a signal	
$P_x = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} \lvert x(t)\rvert^2 dt$	$P_x = \lim_{n\to\infty} \frac{1}{2n+1} \sum_{k=-n}^{n} \lvert x[k]\rvert^2$
Power of a periodic signal:	
$P_x = \frac{1}{T} \int_{t_0}^{t_0+T} \lvert x(t)\rvert^2 dt$	$P_x = \frac{1}{N} \sum_{k=k_0}^{k_0+N-1} \lvert x[k]\rvert^2 dt$
Energy of a signal	
$E_x = \int_{-\infty}^{\infty} \lvert x(t)\rvert^2 dx$	$E_x = \sum_{k=-\infty}^{\infty} \lvert x[k]\rvert^2$

$0 < P_x < \infty$, is called a power signal [unlimited]

$0 < E_x < \infty$, is called a energie signal [limited]

A signal can only be one of the two or none!

[Energie signals have zero Power
& Power signals have Infinity Energie]

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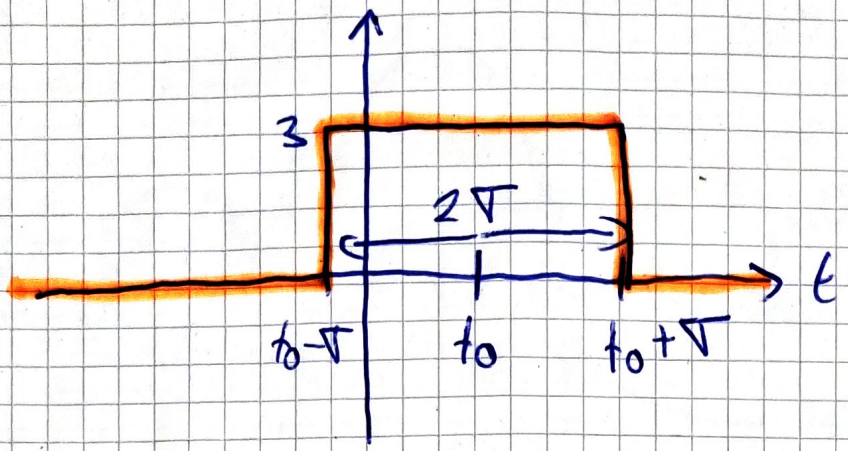
Signale & Systeme Exercise 1

Elementary Signals

a) Define & sketch the shifted & scaled rectangular function

$x(t) = 3\,\mathrm{rect}\left(\frac{t-t_0}{2\tau}\right)$; $t_0 > 0, \tau > 0$



$$x(t) = \begin{cases} 0 & ; \ |t-t_0| > \tau \\ 3/2 & ; \ |t-t_0| = \tau \\ 3 & ; \ |t-t_0| < \tau \end{cases}$$

b) Define one period of a sine function using the periodic sine function $\sin(2\pi f_0 t)$ & the rectangular function

$\sin(2\pi f_0 t) \rightsquigarrow \tau = \frac{2\pi}{2\pi f_0} = \frac{1}{f_0}$

$\mathrm{rect}(f_0 t)$

$x(t) = \mathrm{rect}(f_0 t)\sin(2\pi f_0 t)$

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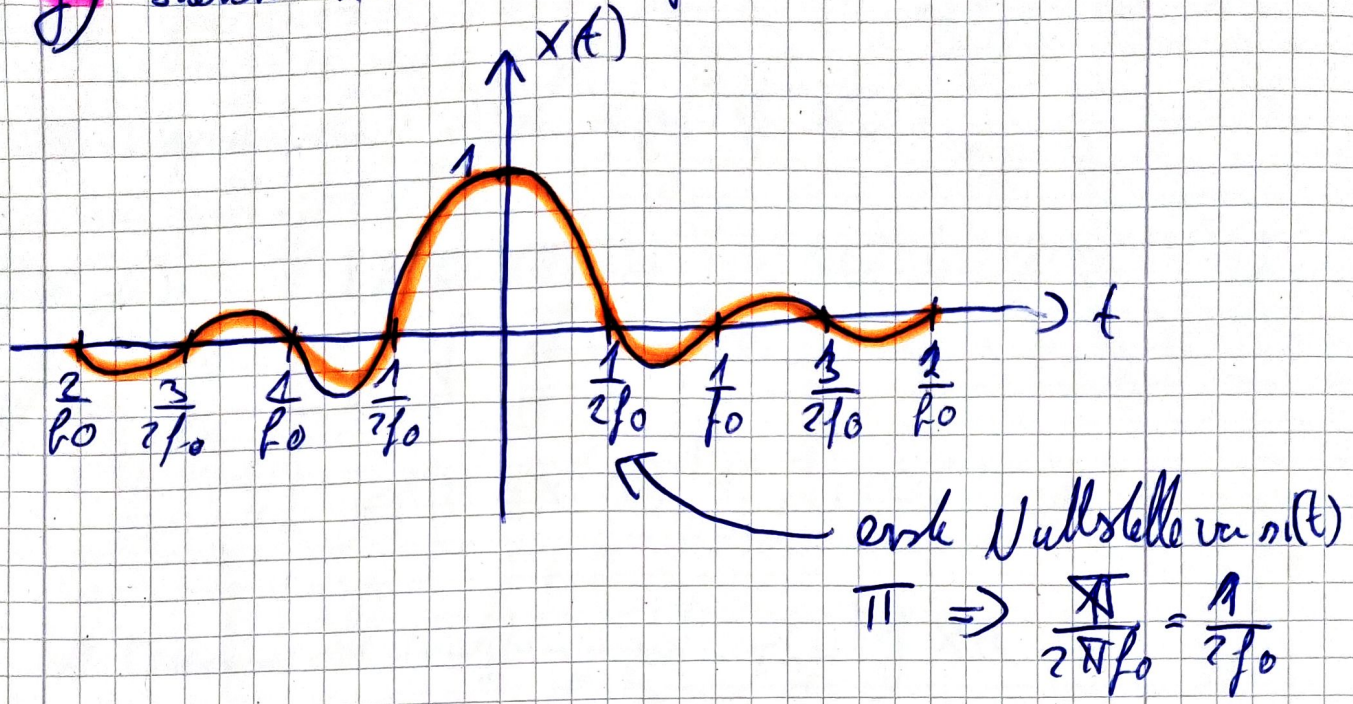
Signale & Systeme

Elementary Signals

1) Sketch $x(t) = si(2\pi f_0 t)$



erste Nullstelle von $si(t)$

$\Pi \Rightarrow \frac{\pi}{2\pi f_0} = \frac{1}{2 f_0}$

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Signale & Systeme

Complex Signals

$$x(t) = 2e^{j2\pi f_0 t} = 2(\cos(2\pi f_0 t) + j\sin(2\pi f_0 t))$$

a) Determine the realpart $x_R(t)$ of $x(t)$

$$x_R(t) = \frac{1}{2}[2\cos(2\pi f_0 t) + j2\sin(2\pi f_0 t) + 2\cos(2\pi f_0 t) - j2\sin(2\pi f_0 t)]$$

$$= \frac{1}{2}[x(t) + x^*(t)] = 2\cos(2\pi f_0 t)$$

b) Determine the imaginary part $x_I(t)$ of $x(t)$

$$x_I(t) = \frac{1}{2j}[x(t) - x^*(t)]$$

$$= \frac{1}{2j}[2\cos(2\pi f_0 t) + j2\sin(2\pi f_0 t) - 2\cos(2\pi f_0 t) + j2\sin(2\pi f_0 t)] = 2\sin(2\pi f_0 t)$$

c) Determine the even part $x_e(t)$ of $x(t)$

$$x_e(t) = \frac{1}{2}[x(t) + x(-t)] = 2\cos(2\pi f_0 t)$$

d) Determine the odd part $x_o(t)$ of $x(t)$

$$x_o(t) = \frac{1}{2}[x(t) - x(-t)] = j2\sin(2\pi f_0 t)$$

e) conjugate even: $x(t) = x^*(-t)$

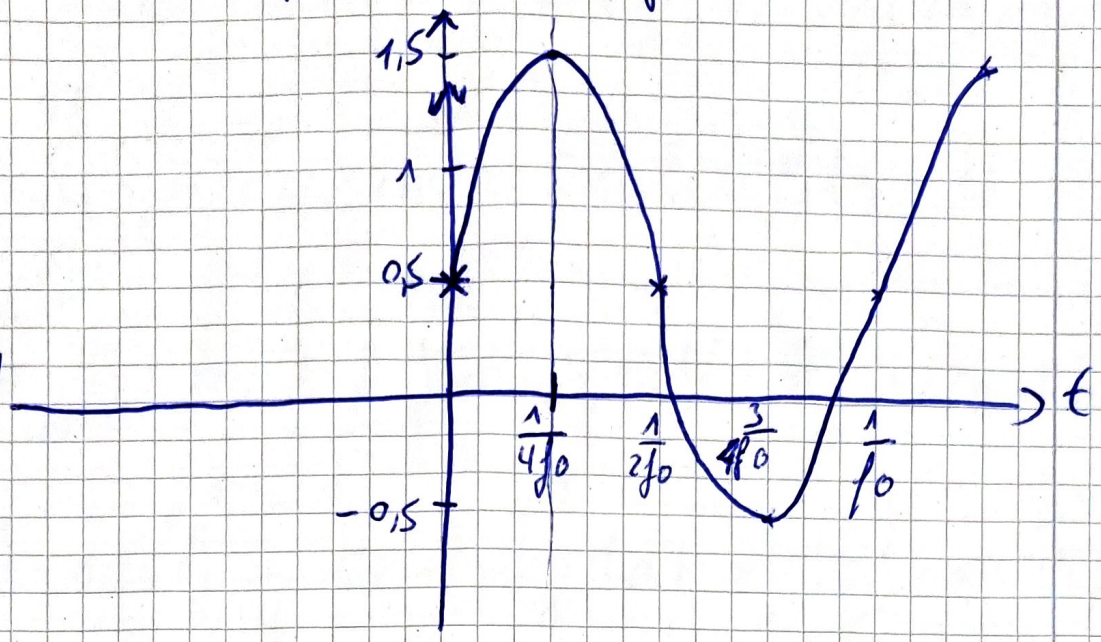
$\cos(t) + j\sin(t) \xrightarrow{*} \cos(t) - j\sin(t) \xrightarrow{-t} \cos(t) + j\sin(t)$

Signale & Systeme

Power & Energy of Signals

$T = \frac{1}{f_0}$

a) $x(t) = 0{,}5 + \sin(2\pi f_0 t)$



|...| braucht man bei complexen Signalen!

$$P_x = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} [0{,}5 + \sin(2\pi f_0 t)]^2 \, dt$$

$$= \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} [0{,}25 + \sin(2\pi f_0 t) + \sin^2(2\pi f_0 t)] \, dt$$

$$= \lim_{T\to\infty} \frac{1}{2T} \left(\int_{-T}^{T} 0{,}25 \, dt + \int_{-T}^{T} \sin(2\pi f_0 t) \, dt + \int_{-T}^{T} \sin^2(2\pi f_0 t) \, dt \right)$$

$\int$ über Periode

$$= \lim_{T\to\infty} \frac{1}{2T} \left(0{,}25 t \Big|_{-T}^{T} + 0 \right)$$

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Power & Energie of signals

$x^* \cdot x$

a)

$$P_x = \frac{1}{T}\int_0^T |x(t)|^2 dt = \frac{1}{T}\int_0^T [0{,}5 + \sin(2\pi f_0 t)]^2 dt$$

$$= \frac{1}{T}\int_0^T [0{,}25 + \sin(2\pi f_0 t) + \sin^2(2\pi f_0 t)]\, dt$$

$$= \frac{1}{T}\int_0^T 0{,}25\, dt + \frac{1}{T}\int_0^T \sin(2\pi f_0 t)\, dt + \frac{1}{2T}\int_0^T (1 - \cos(4\pi f_0 t))\, dt$$

$$= \frac{1}{T} \cdot 0{,}25t\Big|_0^T + 0 + \frac{1}{2T}\int_0^T 1\, dt - \frac{1}{2T}\int_0^T \cos(4\pi f_0 t)\, dt$$

$$= 0{,}25 + \frac{1}{2} = 0{,}75 \Rightarrow \text{Power signal}$$

$\hookrightarrow E_x = \infty$

$$P_{x t_1} = |x(0)|^2 = 0{,}25 + 0 = 0{,}25$$

$$P_x t_2 = |x(1)|^2 = |0{,}25 + \sin(2\pi f_0) + \sin^2(2\pi f_0)|$$

$$\sin^2(at) = \frac{1}{2} - \frac{1}{2}\cos(2at)$$

$\int$ $\int$ (sin $\cdot$ cos) over one period integral $= 0$

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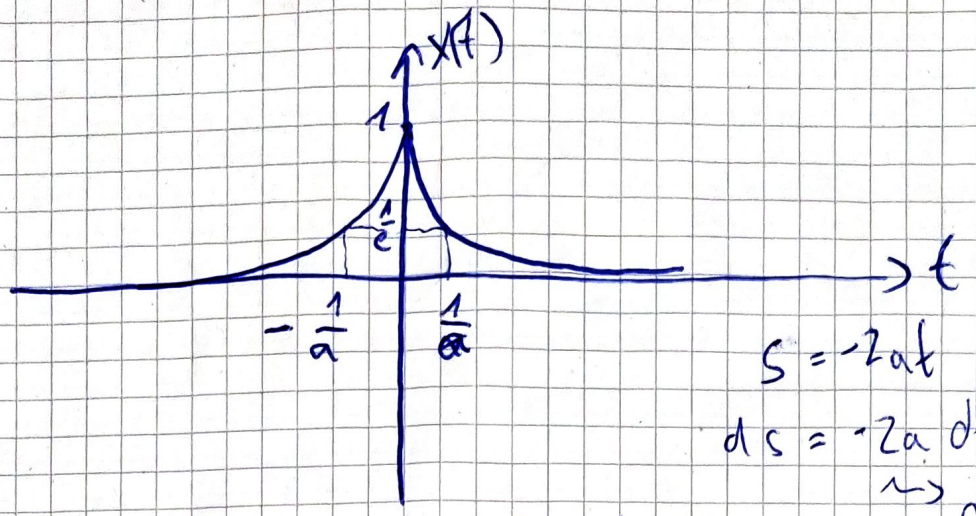
Power & Energy of Signals

b) $x(t) = e^{-|at|}$, $a > 0$



$s = -2at$

$ds = -2a\,dt$

$\rightsquigarrow dt = \frac{ds}{-2a}$

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} e^{-2|at|} dt$$

$$= 2 \cdot \int_0^{\infty} e^{-2at} dt = \frac{2}{-2a} e^{-2at} \Big|_0^{\infty}$$

$$= -\frac{1}{a} e^{-2a \cdot \infty} + \frac{1}{a} e^{-2a \cdot 0} = \frac{1}{a}$$

↳ Energy Signal → $P_x = 0$

$$P_{x\,t_1} = |x(0)|^2 = |e^{-0}|^2 = 1$$

$$P_{x\,t_2} = |x(1)|^2 = |e^{-a}|^2 = e^{-2a}$$

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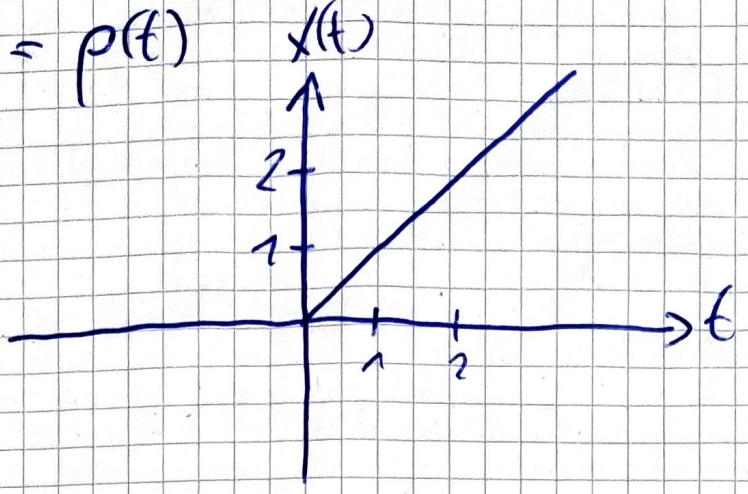
Power & Energy of Signals

d) $x(t) = \rho(t)$



$$P_x = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} |\rho(t)|^2 \, dt$$

$$= \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} \varepsilon(t) x^2 \, dt$$

$$= \lim_{T\to\infty} \frac{1}{2T} \int_{0}^{T} x^2 \, dt = \lim_{T\to\infty} \frac{1}{2T} \left[\frac{x^3}{3}\right]_0^T$$

$$= \lim_{T\to\infty} \frac{T^3}{6T} \to \infty$$

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} \varepsilon(t) x^2 \, dt = \int_{0}^{\infty} x^2 \, dt$$

$$= \left[\frac{x^3}{3}\right]_0^{\infty} \to \infty \quad \to \text{neither power nor energy signal}$$

$$P_{x t_1} = |\rho(0)|^2 = 0$$

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$$P_{x t_2} = |\rho(1)|^2 = 1$$

Signale & Systeme

Orthogonale Systeme

Skalarprodukt of signals

$$\langle x(t), y(t) \rangle = \int_{-\infty}^{\infty} x^*(t) y(t)\, dt$$

$$\langle x[k], y[k] \rangle = \sum_{k=-\infty}^{\infty} x^*[k]\, y[k]$$

Two signals are called orthogonal if

$$\langle x(t), y(t) \rangle = 0$$

$$E_x = \langle x(t), x(t) \rangle$$

Korrelation:

of Energy

Cross correlation function:

$$r_{xy}^E(\tau) = \langle x(t), y(t+\tau) \rangle = \int_{-\infty}^{\infty} x^*(t)\, y(t+\tau)\, dt$$

Autocorrelation function:

$$r_{xx}^E(\tau) = \langle x(t), x(t+\tau) \rangle = \int_{-\infty}^{\infty} x^*(t)\, x(t+\tau)\, dt$$

Energy of a signal:

$$E_x = \langle x(t), x(t) \rangle = r_{xx}^E(0)$$

of Power

Cross correlation function:

$$r_{xy}^P(\tau) = \lim_{\sigma \to \infty} \frac{1}{2\sigma} \int_{-\sigma}^{\sigma} x^*(t)\, y(t+\tau)\, dt \quad // \frac{1}{2n+1} \sum_{k=-n}^{n}$$

Cross correlation for periodic function:

$$r_{xy}^P(\tau) = \frac{1}{\nu} \int_{t_0}^{t_0+\sigma} x^*(t)\, y(t+\tau)\, dt \quad // \frac{1}{N} \sum_{k=k_0}^{k_0+N-1}$$

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$$P_x = r_{xx}^P(0)$$

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Properties of Korrelation

$r_{yx}(\tau) = r_{xy}^*(-\tau)$ // $r_{yx}[k] = r_{xy}^*[-k]$

$r_{xx}(\tau) = r_{xx}^*(-\tau)$ // $r_{xx}[k] = r_{xx}^*[-k]$

↳ Autokorrelation function is conjugate even

$r_{xx}(0) \geq |r_{xx}(\tau)| \Rightarrow \max\{|r_{xx}(\tau)|\} = r_{xx}(0)$

$E_x = r_{xx}^E(0)$ & $P_x = r_{xx}^P(0)$

$\sin(x)\cos(y) = \frac{1}{2}(\sin(x-y) + \sin(x+y))$!

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Signale & Systeme Exercise 2

Orthogonal Signals

b) Show that two signals $s_1(t) = e^{j2\pi k f_0 t}$ and $s_2(t) = e^{j2\pi l f_0 t}$, $k, l \in \mathbb{Z}$ are orthogonal for $k \neq l$ in the interval $[0, 1/f_0]$

$$\langle s_1(t), s_2(t) \rangle = \int_0^{1/f_0} (e^{j2\pi k f_0 t})^* (e^{j2\pi l f_0 t})\, dt$$

$$= \int_0^{1/f_0} (\cos(2\pi k f_0 t) - j\sin(2\pi k f_0 t))(\cos(2\pi l f_0 t) + j\sin(2\pi l f_0 t))\, dt$$

$$= \int_0^{1/f_0} \cos(2\pi k f_0 t)\cos(2\pi l f_0 t) + j\cos(2\pi k f_0 t)\sin(2\pi l f_0 t) - j\sin(2\pi k f_0 t)\cos(2\pi l f_0 t) + \sin(2\pi k f_0 t)\sin(2\pi l f_0 t)\, dt$$

$$= \int_0^{1/f_0} \cos(2\pi(k+l)f_0 t) + j\sin(2\pi(k-l)f_0 t)\, dt$$

↳ $\cos(2\pi n) = 1$, $\cos(0) = 1$ } 1-1

↳ $\sin(2\pi n) = 0$, $\sin(0) = 0$

since $k \neq l$

$$= 0 \rightarrow \text{orthogonal}$$

$$\sin(\alpha \pm \beta) = \sin(\alpha)\cos(\beta) \pm \cos(\alpha)\sin(\beta)$$

$$\cos(\alpha \pm \beta) = \cos(\alpha)\cos(\beta) \mp \sin(\alpha)\sin(\beta)$$

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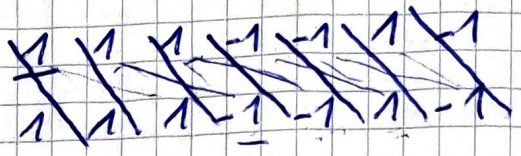
Signale & Systeme

Correlation of Discrete-Time Signals

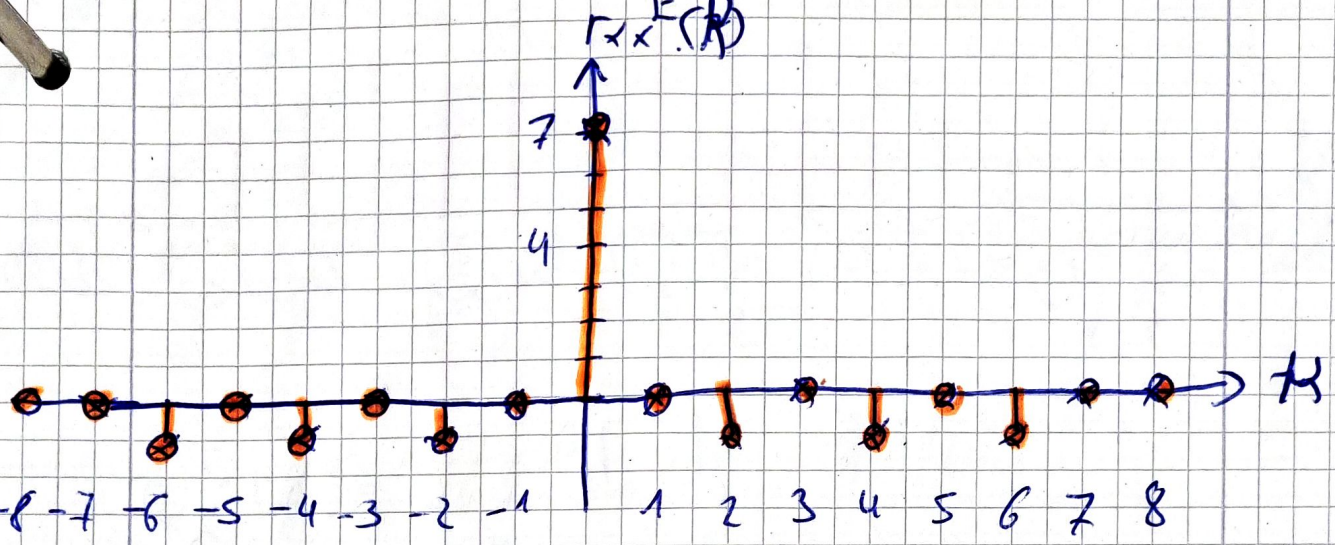
$$x[k] = [1\ 1\ 1\ -1\ -1\ 1\ -1]$$

a)



Fall $K = 1$

$$r_{xx}^{E}[K] = \sum_{k=-\infty}^{\infty} x^*[k] \cdot x[k+K]$$



$$r_{xx}^{E}[K] = \begin{cases} 7, & K = 0 \\ -1, & |K| \in \{2,4,6\} \\ 0, & \text{else} \end{cases}$$

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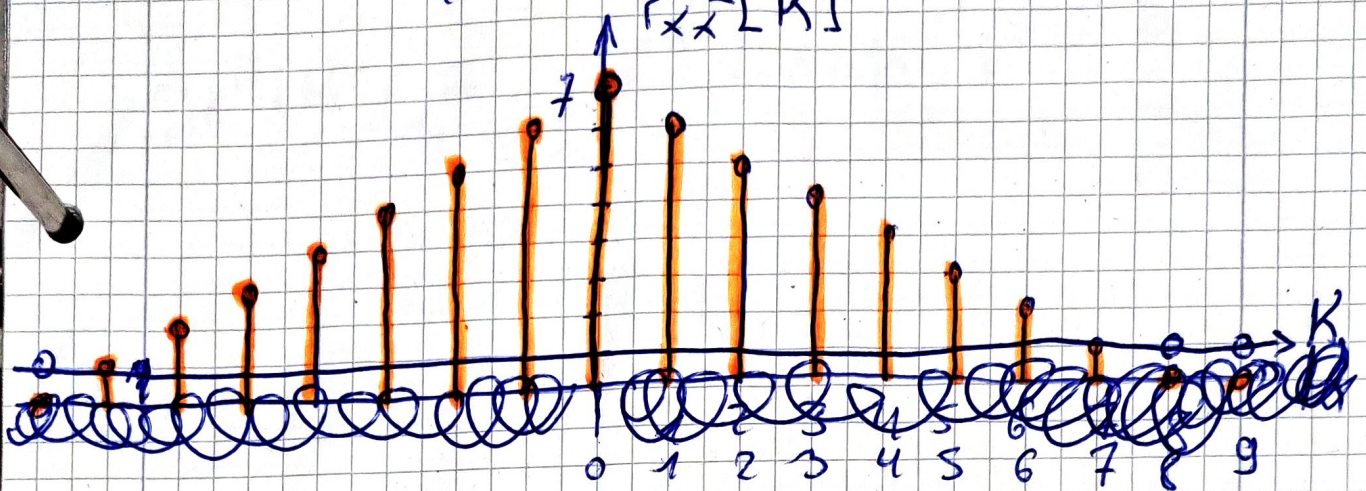
Signale & Systeme

Correlation of Discrete-Time Signals

b) 1 1 1 1 1 1 1
1 1 1 1 1 1 1

$$r_{xx}^{E}[K] = \sum_{k=-\infty}^{\infty} x^*[k]\, x[k+K]$$



$$r_{xx}^{E}[K] = \begin{cases} 7 - |K| & , |K| \in [0,7] \\ 0 & , \text{else} \end{cases}$$

c) For apps like radar or synchronization, the autocorrelation function for $K \neq 0$ should be close to zero such that there is one sharp peak which can be easily detected.
Therefore, the Barker code has much better properties!

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Signals & Systems

Correlation of Discrete Time Signals

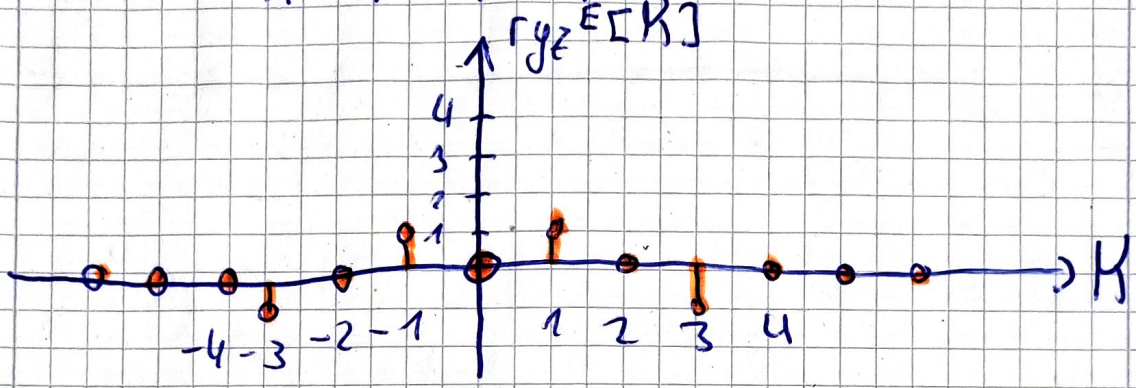
$y[k] = [1 \; -1 \; 1 \; -1] \quad \& \quad x[k] = [1 \; 1 \; -1 \; -1]$

d) determine the cross-correlation function $r_{yx}^{E}[k]$

$$r_{yx}^{E}[k] = \sum_{q=-\infty}^{\infty} x^{*}[q] \, y[q+k]$$

1 -1 1 -1

1 1 -1 -1



$$r_{yx}^{E}[k] = \begin{cases} 1, & k \in \{1, -1\} \\ -1, & k \in \{3, -3\} \\ 0, & \text{else} \end{cases}$$

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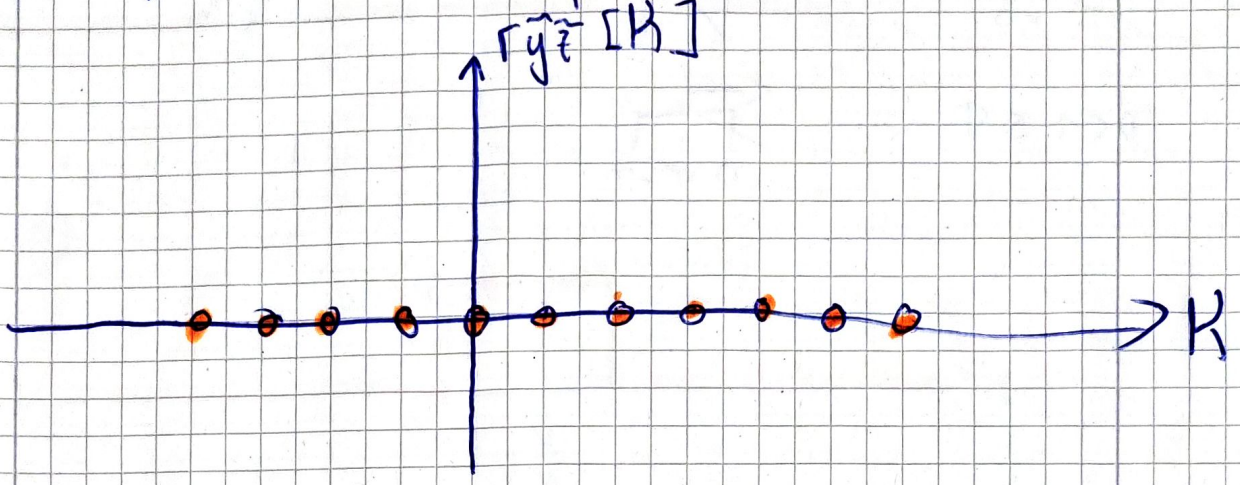
Correlation of Periodic Time Signal:

e) $g[k] \hat{=}$ periodic $g[k]$

$\hat{z}[k] \hat{=}$ periodic $z[k]$

1 -1 1 -1 1 -1 -1 -1

1 -1 -1 -1 1 1 -1 -1



$$r_{\tilde{y}\tilde{z}}^{p}[K] = \frac{1}{N} \sum_{k=0}^{N-1} x^*[k]\, y[k+K] = 0$$

$$\Rightarrow$$

$\tilde{y}[k] \perp \tilde{z}[K]$

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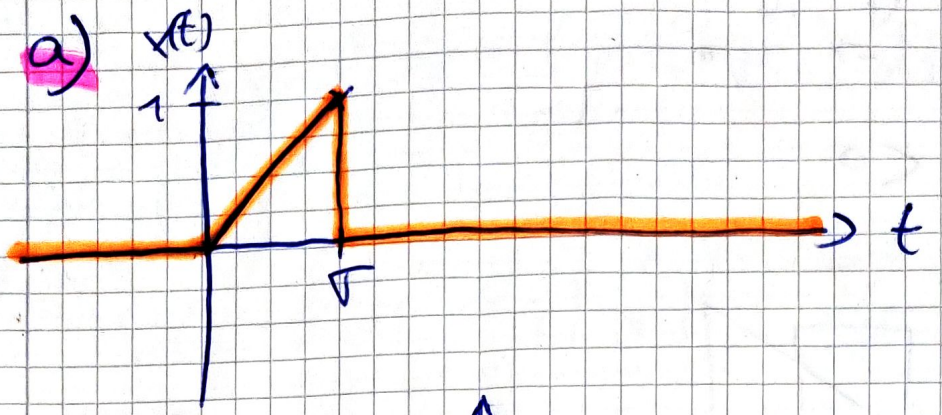
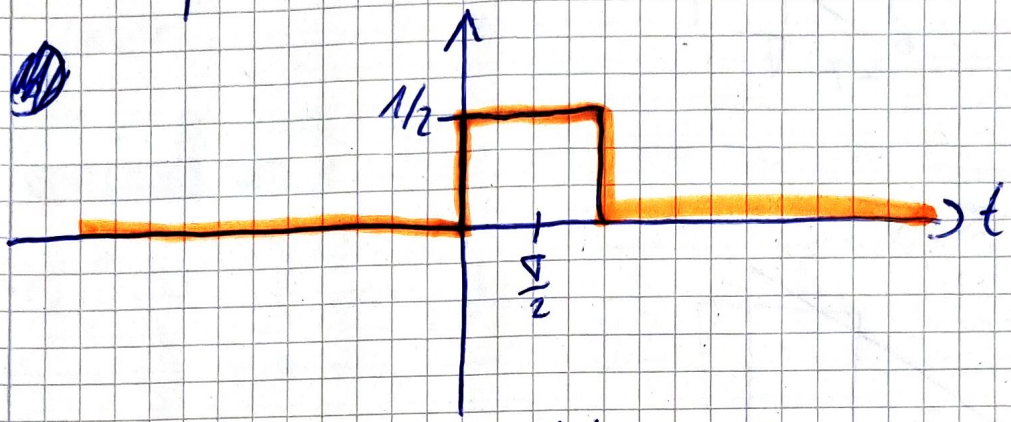
$$x(t) = \begin{cases} \frac{t}{T} & , 0 \le t \le T \\ 0 & , \text{else} \end{cases} \qquad y(t) = \frac{1}{2} \operatorname{rect}\left(\frac{t - T/2}{T}\right)$$

a)



b)

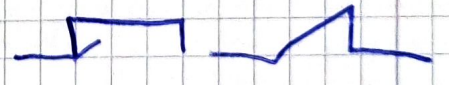


b) Determine the cross correlation $r_{xy}^{E}(\tau)$

$$r_{xy}^{E}(\tau) = \langle x(t), y(t+\tau) \rangle = \int_{-\infty}^{\infty} x^{*}(t)\, y(t+\tau)\, dt$$

$\tau \ge T$: $r_{xy}^{E}(\tau) = 0$

$\tau \le -T$: $r_{xy}^{E}(\tau) = 0$



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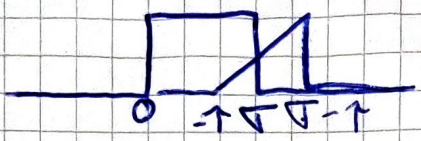
b)



$-T \leq \tau < 0$:

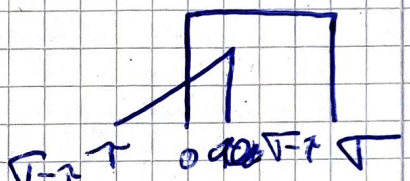
$$r_{xy}^{E}(\tau) = \int_{-\infty}^{\infty} \frac{t}{T} \cdot \frac{1}{2} \operatorname{rect}\left(\frac{t - T/2}{T}\right) dt$$

$$= \int_{-\tau}^{T} \frac{t}{T} \cdot \frac{1}{2}\, dt = \frac{1}{2}\left[\frac{t^2}{2T}\right]_{-\tau}^{T} = \frac{T^2 - \tau^2}{4T}$$

$\tau = 0$:

$$r_{xy}^{E}(\tau) = \int_{0}^{T} \frac{t}{T} \cdot \frac{1}{2}\, dt = \frac{T^2}{4T} = \frac{T}{4}$$

$0 < \tau \leq T$:



$$r_{xy}^{E}(\tau) = \int_{0}^{T-\tau} \frac{t}{T} \cdot \frac{1}{2}\, dt = \left[\frac{t^2}{4T}\right]_{0}^{T-\tau} = \frac{(-\tau + T)^2}{4T}$$

$$r_{xy}^{E}(\tau) = \begin{cases} 0, & |\tau| > T \\ \frac{T^2 - \tau^2}{4T}, & -T \leq \tau < 0 \\ T/4, & \tau = 0 \\ \frac{(-\tau + T)^2}{4T}, & 0 < \tau \leq T \end{cases}$$

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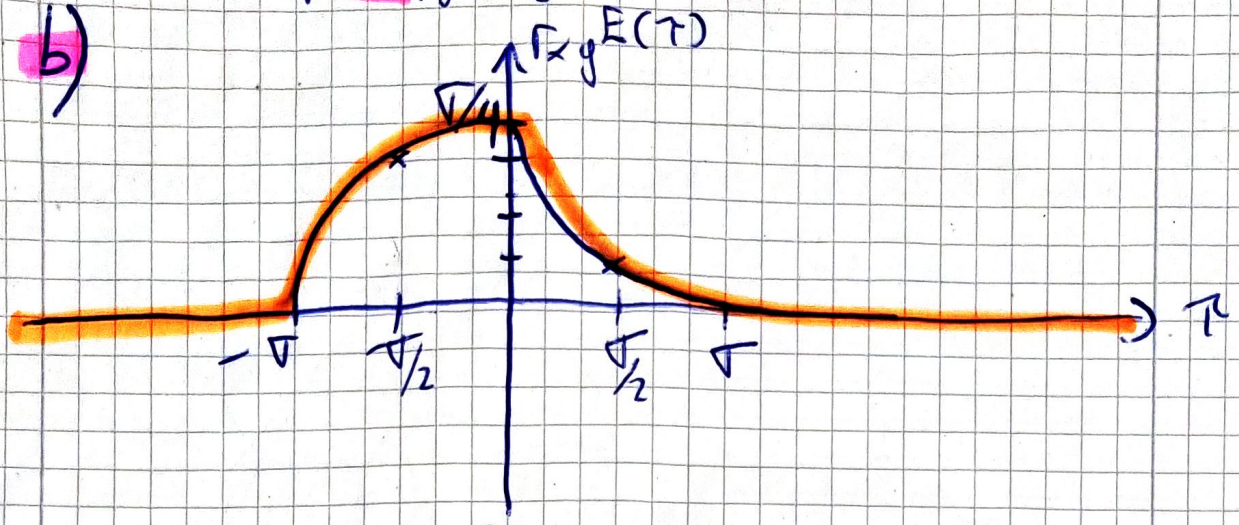
b)



$$r_{xy}^E(T/2) = \frac{T^2 - \frac{T^2}{4}}{4T} = \frac{\frac{3T^2}{4}}{4T} = \frac{3T}{16}$$

$$r_{xy}^E(T/2) = \frac{(T - T/2)^2}{4T} = \frac{\frac{T^2}{4}}{4T} = \frac{1T}{16}$$

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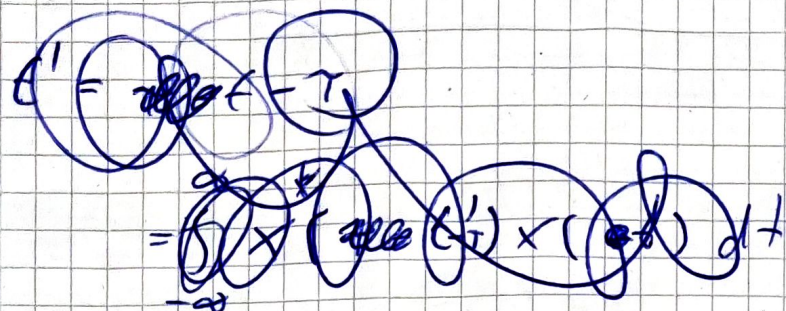
c) Show that $r_{xx}(\tau) = r_{xx}^*(-\tau)$ holds

$$r_{xx}(\tau) \overset{E}{=} \langle x(t), x(t+\tau) \rangle = \int_{-\infty}^{\infty} x^*(t)\, x(t+\tau)\, dt$$



$$r_{xx}^*(-\tau) = \int_{-\infty}^{\infty} x^*(t)\, x(t-\tau)\, dt^*$$

$$= \int_{-\infty}^{\infty} x(t)\, x^*(t-\tau)\, dt = \int_{-\infty}^{\infty} x^*(t')\, x(t'+\tau)\, dt'$$

$t' = t - \tau$

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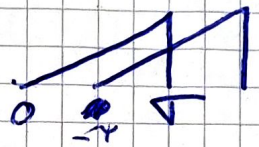
e) Determine $r_{xx}^{E}(\tau)$

$$r_{xx}^{E}(\tau) = \int_{-\infty}^{\infty} x^*(t)\, x(t+\tau)\, dt$$

$\tau \leq -T$: $r_{xx}^{E}(\tau) = 0$

$\tau > T$: $r_{xx}^{E}(\tau) = 0$

$-T \leq \tau < 0$:



$$r_{xx}^{E}(\tau) = \int_{-\tau}^{T} \frac{t}{T} \cdot \frac{t+\tau}{T}\, dt$$

$$= \int_{-\tau}^{T} \frac{t^2 + \tau t}{T^2}\, dt = \left[\frac{t^3}{3T^2} + \frac{\tau t^2}{2T^2} \right]_{-\tau}^{T}$$

$$= \frac{T^3}{3T^2} + \frac{\tau T^2}{2T^2} - \frac{(-\tau)^3}{3T^2} - \frac{\tau(-\tau)^2}{2T^2}$$

$$= \frac{T^3 + \tau^3}{3T^2} + \frac{\tau T^2 - \tau^3}{2T^2} = \frac{2T^3 + 2\tau^3 + 3\tau T^2 - 3\tau^3}{6T^2}$$

$$= \frac{2T^3 + 3\tau T^2 - \tau^3}{6T^2}$$

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e)

$\tau = 0$

$$r_{xx}^{E}(\tau) = \int_0^T \frac{t}{T}\,\frac{t}{T}\,dt = \left[\frac{t^3}{3T^2}\right]_0^T$$

$$= \frac{T^3}{3T^2} = \frac{T}{3}$$

$0 < \tau \leq T$:

$$r_{xx}^{E}(\tau) = \int_0^{T-\tau} \frac{t}{T}\,\frac{t+\tau}{T}\,dt = \int_0^{T-\tau} \frac{t^2+\tau t}{T^2}\,dt$$

$$= \left[\frac{t^3}{3T^2} + \frac{\tau t^2}{2T^2}\right]_0^{T-\tau}$$

$$= \frac{(T-\tau)^3}{3T^2} + \frac{\tau(T-\tau)^2}{2T^2} = \frac{2(T-\tau)^3 + 3\tau(T-\tau)^2}{6T^2}$$

$$r_{xx}^{E}(\tau) = \begin{cases} 0, & |\tau| > T \\ \dfrac{2T^3 + 3\tau T^2 - \tau^3}{6T^2}, & -T \leq \tau < 0 \\ T/3, & \tau = 0 \\ \dfrac{2(T-\tau)^3 + 3\tau(T-\tau)^2}{6T^2}, & 0 < \tau \leq T \end{cases}$$

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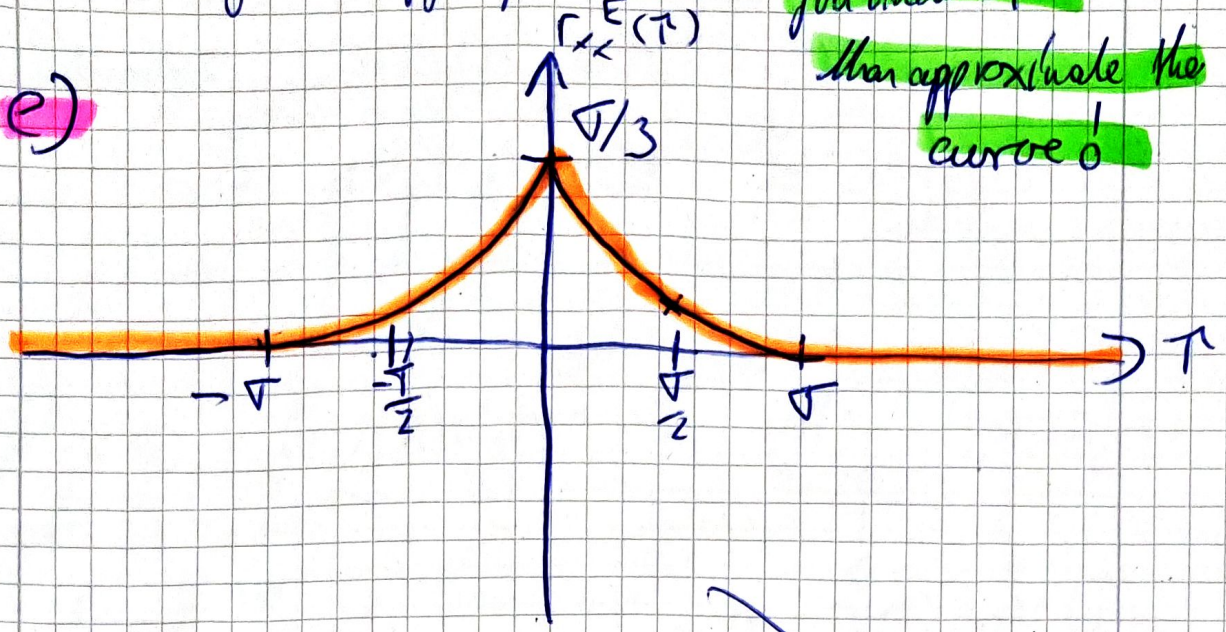
You know $-\tau^3$ that approximate the curve ó

e)



$$r_{xx}^E\left(\frac{\nabla}{2}\right) = \frac{2\left(\nabla-\frac{\nabla}{2}\right)^3 + 3\nabla/2\left(\nabla-\nabla/2\right)^2}{6\nabla^2}$$

$$= \frac{2\left(\nabla/2\right)^3 + \frac{3\nabla}{2}\left(\nabla/2\right)^2}{6\nabla^2} = \frac{\frac{2\nabla^3}{8} + \frac{3\nabla}{2}\frac{\nabla^2}{4}}{6\nabla^2}$$

$$\frac{\frac{2\nabla}{8} + \frac{3\nabla}{8}}{6} = \frac{5\nabla}{\quad}$$

$$r_{xx}^E\left(-\frac{\nabla}{2}\right) = \frac{2\nabla^3 + \frac{3\nabla}{2}\nabla^2 - \left(\frac{\nabla}{2}\right)^3}{6\nabla^2}$$

$$= \frac{2\nabla^3 + \frac{3\nabla^3}{2} - \frac{\nabla^3}{8}}{6} = \frac{\frac{7\nabla}{2} - \frac{\nabla}{8}}{6} =$$

f) $E_x = r_{xx}^E(0) = \nabla/3$

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